

CH12.2 空間向量的應用

$$1. \frac{\sqrt{30}}{30} \quad 2. \left(\frac{1}{2}, -\frac{2}{3}, -\frac{21}{2}\right) \quad 3. (1)(4, -8, 8)$$

$$(2) (6, -8, 11) \quad (3) 18 \quad 4. \left(-2, -1, -\frac{2}{3}\right)$$

$$5. 3, (4, 2, 2) \quad 6. 1(1)(-7, -7, 7) \quad (2) 0$$

$$7. (1) \frac{\sqrt{146}}{2} \quad (2) 6\sqrt{146} \quad 8. (1) -5 \text{ 或 } -1 \quad (2) -3 \quad 9. 120$$

10. -9

解析

1. 設 $C(0,0,0), D(2,0,0), B(0,2,0)$

$$A(2,2,0), G(0,0,2), H(2,0,2)$$

$$F(0,2,2), E(2,2,2)$$

$\therefore P、Q$ 分別為 $\overline{BF}$ 、 $\overline{GH}$ 之中點

$$\therefore P\left(\frac{0+0}{2}, \frac{2+2}{2}, \frac{0+2}{2}\right) = (0, 2, 1)$$

$$Q\left(\frac{0+2}{2}, \frac{0+0}{2}, \frac{2+2}{2}\right) = (1, 0, 2)$$

$$\overrightarrow{PA} = (2, 0, -1), \overrightarrow{PQ} = (1, -2, 1)$$

$$\overrightarrow{PA} \cdot \overrightarrow{PQ} = |\overrightarrow{PA}| |\overrightarrow{PQ}| \cos \theta$$

$$1 = \sqrt{2^2 + 0^2 + (-1)^2} \cdot \sqrt{1^2 + (-2)^2 + 1^2} \cdot \cos \theta$$

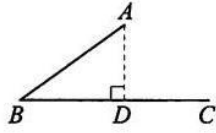
$$\cos \theta = \frac{1}{\sqrt{30}} = \frac{\sqrt{30}}{30}$$

2. $\therefore \vec{a} // \vec{b}$

$$\therefore \frac{1}{3} = \frac{2}{2x+5} = \frac{-1}{3y-1} \Rightarrow \begin{cases} 2x+5=6 \\ 3y-1=-3 \end{cases} \Rightarrow \begin{cases} x=\frac{1}{2} \\ y=-\frac{2}{3} \end{cases}$$

$$\because \vec{b} \perp \vec{c} \therefore (3, 2x + 5, 3y - 1) \cdot (z, 5, x - 1) = 0 \Leftrightarrow (3, 6, -3) \cdot \left(z, 5, -\frac{1}{2}\right) = 0 \Leftrightarrow$$

$$3z + 30 + \frac{3}{2} = 0 \Rightarrow z = -\frac{21}{2}, \text{故序組 } (x, y, z) = \left(\frac{1}{2}, \frac{2}{3}, -\frac{21}{2}\right)$$



3. (1) 設 A 在 $\overrightarrow{BC}$ 上之投影點為 $D(a, b, c)$ ，則 $\overrightarrow{BA}$ 在 $\overrightarrow{BC}$ 上之正射影為 $\overrightarrow{BD}$

$$\text{又 } \overrightarrow{BA} = (8, -10, 4) \quad \overrightarrow{BC} = (2, -4, 4) \Rightarrow \overrightarrow{BD} = \left(\frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BC}|^2} \right) \overrightarrow{BC} =$$

$$\frac{(8, -10, 4) \cdot (2, -4, 4)}{|\sqrt{2^2 + (-4)^2 + 4^2}|^2} (2, -4, 4) = (4, -8, 8)$$

$$(2) \text{承}(1), \overrightarrow{BD} = (a - 2, b - 0, c - 3) = (4, -8, 8) \Rightarrow (a, b, c) = (6, -8, 11)$$

故 A 在 $\overrightarrow{BC}$ 上之投影坐標為 $(6, -8, 11)$

$$(3) \overrightarrow{BA} \times \overrightarrow{BC} = \left(\begin{vmatrix} -10 & 4 \\ -4 & 4 \end{vmatrix}, \begin{vmatrix} 4 & 8 \\ 2 & 2 \end{vmatrix}, \begin{vmatrix} 8 & -10 \\ 2 & -4 \end{vmatrix} \right) \\ = (-24, -24, -12)$$

$$\Delta ABC \text{ 面積} = \frac{1}{2} \cdot \sqrt{(-24)^2 + (-24)^2 + (-12)^2} = \underline{18}$$

$$4. \text{由柯西不等式 } [(2x)^2 + (2y)^2 + (3z)^2](2^2 + 1^2 + 1^2) \geq$$

$$(4x + 2y + 3z)^2$$

$$\Rightarrow 24 \cdot 6 \geq (4x + 2y + 3z)^2 \Rightarrow 12 \geq 4x + 2y + 3z \geq -12$$

$$\text{等號成立} \Leftrightarrow \frac{2x}{2} = \frac{2y}{1} = \frac{3z}{1}$$

即當 $x = 2y = 3z$ 時, $4x + 2y + 3z$ 有最小值為 -12

$$\text{設 } x = t, y = \frac{t}{2}, z = \frac{t}{3}, \text{代入 } 4x + 2y + 3z = -12$$

$$\text{得 } t = -2$$

故當 $(x, y, z) = \left(-2, -1, -\frac{2}{3}\right)$ 時, $4x + 2y + 3z$

有最小值為 -12

5. 令 $\overrightarrow{OP} = (2t, t, t)$

$$\begin{aligned}\overrightarrow{PA} \cdot \overrightarrow{PB} &= (5 - 2t, 4 - t, -1 - t) \cdot (4 - 2t, 2 - t, 1 - t) \\ &= 20 - 18t + 4t^2 + 8 - 6t + t^2 - 1 + t^2 = 6t^2 - 24t + 27 \\ &= 6(t^2 - 4t + 4) + 3 = 6(t - 2)^2 + 3\end{aligned}$$

$\therefore$ 當 $t = 2$ 時, $\overrightarrow{PA} \cdot \overrightarrow{PB}$ 有最小值為 3, 此時 $P(4, 2, 2)$

$$\begin{aligned}6. (1) \overrightarrow{OA} \times \overrightarrow{OB} &= \left(\begin{vmatrix} -1 & 2 \\ 2 & 3 \end{vmatrix}, \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix}, \begin{vmatrix} 3 & -1 \\ 1 & 2 \end{vmatrix} \right) \\ &= (-7, -7, 7)\end{aligned}$$

$$\begin{aligned}(2) \overrightarrow{OA} \cdot (\overrightarrow{OA} \times \overrightarrow{OB}) &= (3, -1, 2) \cdot (-7, -7, 7) \\ &= -21 + 7 + 14 = 0\end{aligned}$$

7. (1) $\overrightarrow{AB} = (1, 1, -4), \overrightarrow{AC} = (-2, 2, -1)$

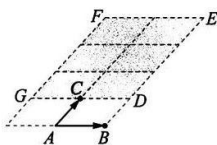
$$\begin{aligned}\overrightarrow{AB} \times \overrightarrow{AC} &= \left(\begin{vmatrix} 1 & -4 \\ 2 & -1 \end{vmatrix}, \begin{vmatrix} -4 & 1 \\ -1 & -2 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ -2 & 2 \end{vmatrix} \right) \\ &= (7, 9, 4)\end{aligned}$$

$\therefore \triangle ABC$ 面積為 $\frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}|$

$$= \frac{1}{2} \sqrt{7^2 + 9^2 + 4^2} = \frac{\sqrt{146}}{2}$$

(2) $\overrightarrow{AP} = r\overrightarrow{AB} + t\overrightarrow{AC}, -1 \leq r \leq 1, 1 \leq t \leq 3$

圖形如右: 所有 P 點所形成之圖形即為平行四邊形 DEFG, 其面積為平行四邊形 ABDC 之 6 倍又平行四邊形 ABDC 面積 $= 2 \cdot \triangle ABC$ 面積 $= \sqrt{146}$ 故所求為 $6\sqrt{146}$



$$8. \overrightarrow{AB} = (-2, 0, -2), \overrightarrow{AC} = (1, -3, -1)$$

$$\overrightarrow{AD} = (a-1, -a-3, -4)$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \left(\begin{vmatrix} 0 & -2 \\ -3 & -1 \end{vmatrix}, \begin{vmatrix} -2 & -2 \\ -1 & 1 \end{vmatrix}, \begin{vmatrix} -2 & 0 \\ 1 & -3 \end{vmatrix} \right)$$

$$= (-6, -4, 6)$$

$$\text{平行六面體體積 } V = |\overrightarrow{AD} \cdot (\overrightarrow{AB} \times \overrightarrow{AC})|$$

$$= |(a-1, -a-3, -4) \cdot (-6, -4, 6)|$$

$$= |-6a + 6 + 4a + 12 - 24| = |-2a - 6| \quad (1) \quad V = |-2a - 6| = 4 \Rightarrow -2a - 6 = \pm 4 \Rightarrow a = -5 \text{ 或 } -1 \quad (2) \text{四點共面} \Rightarrow \text{體積 } V = 0 \Rightarrow |-2a - 6| = 0 \Rightarrow a = -3$$

$$9. \text{依題意, } |(\vec{a} \times \vec{b}) \cdot \vec{c}| = 10, \text{ 又 } \vec{a} \times \vec{a} = 0$$

$$\therefore \vec{a} \times (2\vec{a} + 3\vec{b}) = 3(\vec{a} \times \vec{b})$$

$$\text{所求} = |(\vec{a} \times (2\vec{a} + 3\vec{b})) \cdot (\vec{a} - \vec{b} + 4\vec{c})|$$

$$= |3(\vec{a} \times \vec{b}) \cdot (\vec{a} - \vec{b} + 4\vec{c})|$$

$$= 12|(\vec{a} \times \vec{b}) \cdot \vec{c}| = 120$$

$$10. \begin{vmatrix} b+2c & c+a & a+b \\ y+2z & z+x & x+y \\ q+2r & r+p & p+q \end{vmatrix} = \begin{vmatrix} 3c & c+a & a+b \\ 3z & z+x & x+y \\ 3r & r+p & p+q \end{vmatrix} = \begin{vmatrix} 3c & a & a+b \\ 3z & x & x+y \\ 3r & p & p+q \end{vmatrix}$$

$$\hat{1} \cong \frac{\times 1}{\times (-1)} \frac{+1}{\times \left(-\frac{1}{3}\right)} \frac{\times (-1)}{\times (-1)}$$

$$= \begin{vmatrix} 3c & a & b \\ 3z & x & y \\ 3r & p & q \end{vmatrix} = 3 \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = 3 \times (-3) = \underline{\underline{-9}}$$